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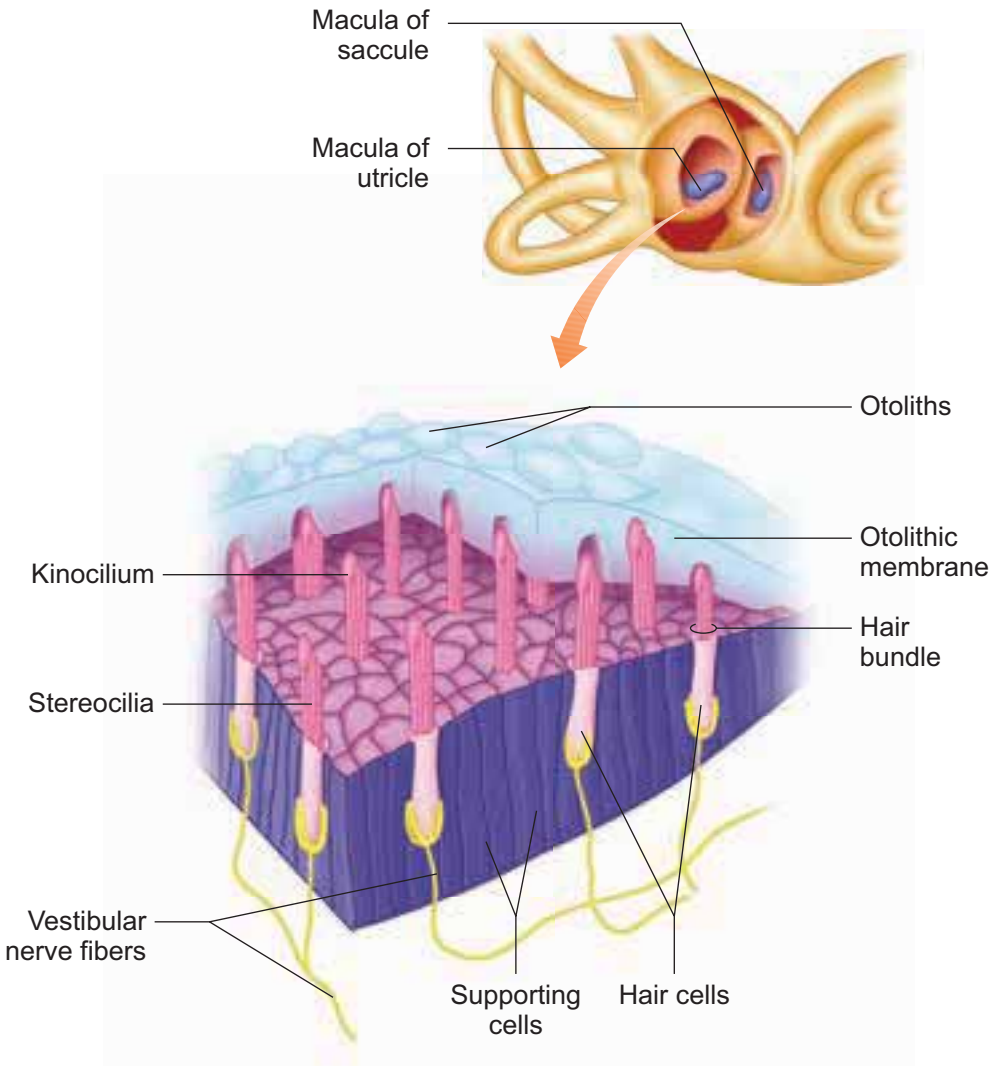


Figure 10.21 Structure of Macula

the position, orientation and movement of the body. The organ of balance is known as the **vestibular system** which is located in the inner ear next to the cochlea. The vestibular system is composed of a series of fluid filled sacs and tubules.These sacs and tubules contain endolymph and are kept in the surrounding perilymph (Figure 10.20). These two fluids, perilymph and endolymph, respond to the mechanical forces, during changes occurring in body position and acceleration (Figure 10.21).

The utricle and saccule are two membranous sacs, found nearest the cochlea and contain equilibrium receptor regions called **maculae** that are involved in detecting the linear movement of the head. The maculae contain the hair cells that act as mechanorecptors. These hair cells are embeded in a gelatinous otolithic membrane that contains small calcareous particles called **otoliths**. This membrane adds weight to the top of the hair cells and increase the inertia.

The canals that lie posterior and lateral to the vestibule are semicircular canals; they are **anterior, posterior and lateral canals** oriented at right angles to each other**.** At one end of each semicircular canal, at its lower end has a swollen area called **ampulla**. Each ampulla has a sensory area known as **crista ampullaris** which is